

First-Principles Framework for Motive-Stabilized Hodge Classes

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Abstract

We propose a unified framework for understanding Hodge classes from first principles. This structure synthesizes insights from K-theory, motive theory, spectral triples, and Tannakian formalism. We define Hodge classes as fixed points of a recursive stabilization process operating on a noncommutative tower of rungs, with realization functors landing in categories of cohomological representations. This gives rise to a generative rather than diagnostic interpretation of the Hodge Conjecture.

1 Layered Framework Overview

1. Primordial Algebra: K-theory

Let $K_0(X)$ denote the Grothendieck group of coherent sheaves or vector bundles on a smooth projective variety X . This forms the base structure:

$$K_0(X) = \{[E] \mid E \text{ coherent sheaf}\}$$

serving as the additive generator space for all higher constructions.

2. Motive Triples and Correspondence Flow

Define pure motives via Grothendieck's triples:

$$M = (X, p, m)$$

where $p \in \text{Corr}^0(X, X)$ is an idempotent correspondence and $m \in \mathbb{Z}$ is a weight shift. Morphisms between motives are given by correspondences of degree $n - m$, allowing filtered interactions among geometries.

3. Recursive Stabilization Operator

Define the operator:

$$T := T_2 \circ T_1$$

where:

- T_1 resolves patching inconsistencies (via MayerVietoris corrections)
- T_2 symmetrizes through canonical correspondences (projectors)

Fixed points under T define stable cycles:

$$\text{StableCycle}^p(X) := \{Z \in Z^p(X) \mid T(Z) = Z\}$$

4. Realization Systems

Following Deligne, each motive M has realizations:

$$(M_B, M_{\text{dR}}, M_{\text{ét}}, M_{\text{cris}})$$

with comparison isomorphisms:

$$\text{comp}_{\text{dR},B}, \quad \text{comp}_{\text{ét},B}, \quad \text{comp}_{\text{cris},\text{dR}}$$

These encode how the same underlying class behaves in different cohomological regimes.

5. Spectral Triple Dynamics

Define a spectral triple (A, H, D) :

- A : algebra of correspondences
- H : Hilbert space of realization data
- D : Dirac-type operator driving evolution

Zeta functions and heat kernels of D capture recursion dynamics:

$$\zeta_D(s) := \text{Tr}(|D|^{-s}), \quad e^{-tD^2}$$

Stabilizing eigenmodes of D correspond to potential Hodge classes.

6. Noncommutative Rung Tower

Define a tower of graded rungs $\mathcal{R}_i$, each with a local algebra A_i . If A_i is noncommutative, then:

$$[T_1, T_2] \neq 0 \quad \Rightarrow \quad \text{obstruction class } \tau \in H^{2p+1}(X)$$

Stability requires respecting rung-level composition constraints.

7. Tannakian Realization and Fixed Points

Let $\omega : \text{Mot} \rightarrow \text{Vect}_{\mathbb{Q}}$ be a neutral fiber functor. The MumfordTate group $\text{MT}(M)$ acts on $\omega(M)$. Then:

$$\gamma \in \text{Inv}_{\text{MT}(M)}(\omega(M)) \quad \Rightarrow \quad \gamma \text{ is a rational Hodge class}$$

2 Conclusion: The First Principle

A Hodge class arises as a fixed tensor:

- Defined recursively from K-theory generators
- Stabilized through motive correspondences
- Converging under spectral recursion
- Realized compatibly in all cohomological frames
- Invariant under Tannakian symmetry

This recasts the Hodge Conjecture as a statement of categorical and spectral fixed-point stability within a motive-theoretic framework.